

Formula

$$\bar{d} \pm t_{\{\alpha/2, n-1\}} \cdot \frac{s_d}{\sqrt{n}}$$

\$ \$

where $\bar{d}$ is the mean difference and s_d is the standard

deviation of differences.

Example

Heart-rate data: $\bar{d} = 6.1$, $s_d = 4.0$, $n = 12 \Rightarrow 95\% \text{ CI} = 6.1$

$$\pm 2.201 \cdot 4.0 / \sqrt{12} \Rightarrow (3.6, 8.6).$$

Interpretation

If 0 lies outside the interval, the two-tailed paired t-test

at α matches the CI decision (reject H_0 when 0 not in CI).